

Homework 0

Restaurant Scenario

- Two intelligent agents working in roles of restaurant employees
- **Servicer agent** is primarily responsible for tasks such as servicing guests and seating management
- **Kitchen agent** is primarily responsible for tasks such as managing orders, cooking, and kitchen inventory

<Goal>

Servicer: propose make potato dish

//Kitchen checks inventory, we have run out of potatoes//

Kitchen: rejects making potato dish

Servicer: propose make steak dish instead

//Kitchen checks inventory, steak is in stock//

Kitchen: accepts making steak dish

<Plan>

Servicer: propose plan to develop a new seasonal menu as it is now winter

Kitchen: refine plan - how many items should we include on the seasonal menu?

Servicer: estimate of ten different dishes on the menu

Kitchen: reject plan - our stocking rate for such a variety of ingredients can only accommodate providing at most seven dishes on the menu at a time

Kitchen: modify and accept new plan - develop a new seasonal menu with seven dishes

<Commitment>

****Note: define commitment as social commitment of one agent to another agent that they will completely commit to doing a task****

Servicer: propose commitment of Kitchen agent, please make sure to stock up on pufferfish for the weekend.

Kitchen: reject commitment - I will try to stock up on pufferfish, but because it is not in season right now I cannot commit that I will be able to obtain it for this weekend

Servicer: propose commitment of Kitchen agent, please make sure to stock up on sweet potatoes in time for this weekend.

Kitchen: accept commitment - Okay, I guarantee I will make sure we have sweet potatoes this weekend. I have already contacted the vendor and the order should be on its way.

<Goal breakdown>

Servicer: propose - goal, we should try to make guests more satisfied with the restaurant.

Kitchen: reject initial proposal

Kitchen: refine - we should identify more specific ways of improving the restaurant like service or food quality to make guests more satisfied

Servicer: modify proposal - we will aim to improve aspects of the restaurant such as lowering waiting time and making food even more palatable to guests

Kitchen: query - by what methods will we lower waiting time and make food more palatable to guests?

Servicer: inform - we may attempt to involve another Servicer or Kitchen agent to service guests at a faster speed to lower waiting time, and observe a pattern amongst the most frequently ordered dishes on the menu to determine what types of flavors appear to be more popular, and adjust the menu to include more similar dishes

Kitchen: accept modified and specified proposal - the goal of making guests happier has been distributed into more detailed and specific goals designed to achieve our operationalization of guest "satisfaction"

<Organization structure>

Note: the "theme" of this organization structure topic regards the two IAs discussing how they should organize themselves regarding restocking of cooking ingredients in the restaurant

Kitchen: propose - should I automatically restock each ingredient as it reaches some predefined level that we determine together ahead of time?

Servicer: refine - you should also take into account dynamically what kinds of ingredients may be running low in particular based on the kinds of dishes guests have been ordering most recently

Kitchen: modify proposal - By default I will automatically restock each ingredient as it reaches some predefined threshold that we determined together ahead of time, but if you (Servicer) find that there is a significant change in the kinds of dishes guests are ordering from our known historical distribution of dishes ordered you (Servicer) should notify me and I will update the threshold used to determine when to restock each item.

Servicer: accept modified proposal - yes this sufficiently accounts for the point that I brought up

<Schedule>

Servicer: propose open restaurant at 6:00AM

Kitchen: reject - must trade with daily fresh fish vendor who opens at 5:30AM, so will not be able to prepare for opening in time by 6:00AM

Kitchen: modify - let's open at 6:30AM that will give me enough time to prepare

Servicer: accept Kitchen proposal to open at 6:30AM

Kitchen: refine - should the schedule include a closing time in the middle of the day so we can prepare for dinner service?

Servicer: yes, that would be appropriate

Kitchen: query - what is the restaurant closing time for the proposed schedule?

Servicer: inform - closing time is set to be 10:30PM

<Task assignment>

Kitchen: propose assignment of task - Servicer can you retrieve produce for dinner service during the lunch service as I will be busier with other tasks during that time

Servicer: reject assignment - I will be fully occupied with guests during lunch service and will not be able to retrieve produce

Kitchen: propose assignment of task - Servicer can you retrieve produce for dinner service during the mid-day closing time

Servicer: refine assignment - which selection of the produce do you want me to retrieve?

Kitchen: all of it please

Servicer: modify assignment - but some of the produce is not used during the dinner service and retrieving it prematurely will cause wilting and render the rest of it unusable. I will only retrieve the produce that will be used during dinner service

Kitchen: accept task proposal modified by Servicer

Servicer: self-accept task assigned to it

<Resource allocation>

****Note: theme of resource allocation is allocating some furniture needed for interior renovation of the restaurant****

Servicer: propose - let us allocate some new chairs and tables to accommodate the increasing number of guests that have been frequenting the restaurant

Kitchen: refine - we should specify how many chairs and tables we will be ordering this time

Servicer: modify proposal to be more detailed - we should order 5 tables and 20 chairs to accommodate the average amount of guests that have been waiting in line for this year's history

Kitchen: reject proposal - the interior space of the restaurant can only accommodate at most 4 more tables and still adhere to safety law

Servicer: accept proposal - we should order 4 tables and 16 chairs

Kitchen: accept proposal

<Untyped info>

Kitchen: query - Servicer do you think that the guests have a positive impression of the restaurant?

Servicer: inform - Yes, I often receive positive remarks from guests like "The food tasted good" or "Service was frequent and unintrusive"